

Notes: Fitzgerald and Haller (RES, 2014)

Pricing-to-Market: Evidence From Plant-Level Prices

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1 Big picture

This note summarizes Fitzgerald and Haller (2014), which uses matched Irish plant-level producer price data to study whether firms price the same product differently across home and foreign markets, and whether this behavior survives after conditioning on actual price changes. The central question is not simply whether export prices are sticky in local currency. Instead, the paper asks whether, when firms reset prices, they choose foreign-market markups that move systematically with the nominal exchange rate.

One-sentence summary. The paper combines monthly plant-product-level prices for Irish domestic sales and UK exports with invoice currency information to show that, for Sterling-invoiced exports, the foreign-to-home markup ratio moves almost one-for-one with the Irish pound/euro-Sterling exchange rate even conditional on synchronized price changes, implying a strong form of pricing-to-market that is not merely mechanical price stickiness.

Core result. When a producer sells the same product in Ireland and in the UK, and UK prices are invoiced in Sterling, the estimated coefficient on the exchange rate in the intensive-margin reset-price equation is about 1.01, with standard error 0.09. Thus, when the home currency depreciates against Sterling, the home-currency value of the UK reset price rises roughly one-for-one relative to the Irish reset price. Under constant relative marginal cost, this means the UK markup rises one-for-one relative to the Irish markup.

Why the result matters. Much of the older pricing-to-market literature could not cleanly separate true markup adjustment from price stickiness, local content, destination-market distribution costs, or product heterogeneity. Fitzgerald and Haller make progress because they observe producer prices at the factory gate, the same plant, the same product category, both domestic and export prices, invoice currency, and monthly timing of price changes.

2 Incremental contribution of the paper

First, the paper improves measurement. Instead of comparing aggregate export unit values across destinations, the authors compare prices charged by the same Irish plant for the same product category in the domestic market and the UK export market. This reduces the product-composition and firm-composition problems that complicate standard trade-price regressions.

Second, the paper isolates pricing-to-market beyond sticky prices. A nominal exchange rate change mechanically alters the home-currency value of a Sterling price if the Sterling price is unchanged. The paper therefore conditions on episodes in which prices actually change. The object of interest is the reset price, not the stale price.

Third, the paper uses invoice currency as a key identifying dimension. The main sample focuses on Sterling-invoiced export prices because Sterling invoicing strongly identifies the UK destination and represents local-currency pricing for Irish exporters selling to the UK. The paper also studies a smaller home-currency invoicing subsample and finds much weaker pricing-to-market behavior.

Fourth, the paper provides broad evidence across manufacturing. Many structural pricing-to-market papers study a specific good or industry, such as coffee, beer, cars, or super-

market products. This paper covers a wide range of Irish manufacturing plants and sectors, so the result is not limited to one special market.

Fifth, the paper offers a disciplined empirical moment for international macro models.

The finding that reset prices themselves move with nominal exchange rates under local-currency invoicing suggests that real exchange-rate volatility and persistence may reflect not only the mechanical effect of nominal rigidity but also the way firms choose new prices when they do adjust.

3 Research question and economic mechanism

The paper asks whether firms adjust relative markups across markets when the exchange rate changes. In the notation of the paper, the home-currency price charged by plant-product pair i in market k can be written as

$$P_t^{ik} = E_t^k P_t^{ik*} = \mu_t^{ik} (1 + \tau^{ik}) C_t^i, \quad (1)$$

where E_t^k is the exchange rate, P_t^{ik*} is the local-currency destination-market price, C_t^i is marginal cost in home currency, τ^{ik} is a time-invariant market service cost, and μ_t^{ik} is the markup.

There is pricing-to-market if

$$\frac{\mu_t^{ik}}{\mu_t^{iH}} \quad (2)$$

varies over time, especially if it co-moves with the nominal exchange rate. The paper focuses on whether this relative markup movement persists conditional on price changes.

The economic mechanism is as follows. Under local-currency pricing, if the Sterling price is unchanged, a depreciation of the home currency mechanically raises the home-currency value of the UK price. But the crucial finding is stronger: even when the firm changes its prices in both the home and UK markets, the firm still does not fully undo this exchange-rate-induced markup movement. Instead, it resets prices in a way that lets relative markups continue to move with the exchange rate.

4 Data and institutional setting

4.1 Data sources

The paper merges two confidential Irish data sources.

Irish Census of Industrial Production (CIP). This is an annual plant census covering manufacturing, mining, and utilities. It contains plant identifiers, industrial classification, ownership, sales, exports, employment, wage bills, materials, energy, import information, export destination information, and invoice-currency information.

Irish Producer Price Index / Wholesale Price Index microdata. This is a monthly price survey. Plants report prices for their main products. The key feature is that domestic and export prices are collected in the same survey, and plants are encouraged to provide parallel home and export price quotes for products sold in both markets.

4.2 Main empirical sample

The main sample focuses on plant-product pairs with a domestic price quote and a Sterling-invoiced UK export price quote. This is useful because Sterling-invoiced exports are highly likely to be UK sales, and they represent local-currency invoicing from the perspective of Irish producers.

The relevant sample period is mostly January 1995 to December 2005, with price data available through November 2006. The period includes a large depreciation of the Irish pound/euro against Sterling followed by an appreciation, which provides meaningful exchange-rate variation.

4.3 Why Ireland and the UK are a useful setting

Ireland and the UK are geographically close and economically integrated, but their currencies moved against each other over the sample period. Irish plants often sell the same product to both markets. Because the price data are factory-gate prices, the analysis is less contaminated by downstream retailing, distribution margins, and local-content costs than consumer price data.

5 Model and empirical design

5.1 Reset prices

Let $\hat{P}_t^{ik*}$ be the local-currency reset price chosen by plant-product pair i in market k when it changes price. The reset price is observed only in periods when the price changes. The paper approximates the home-currency log reset price by

$$\hat{p}_t^{ik} = \alpha^{ik} + c_t^i + \beta_e^k e_t^k + \varepsilon_t^{ik}, \quad (3)$$

where c_t^i captures all plant-product-time effects common across markets, including marginal cost; e_t^k is the exchange rate; and β_e^k captures how the markup in market k responds to exchange-rate movements.

For the UK market under Sterling invoicing, the key parameter is β_e^{UK} . If $\beta_e^{UK} = 1$, the home-currency reset price in the UK moves one-for-one with the exchange rate relative to the Irish reset price, implying that relative markups move one-for-one with the exchange rate. If $\beta_e^{UK} = 0$, relative markups do not respond to exchange-rate movements.

5.2 Intensive-margin specification

The intensive margin studies price changes when prices change in both markets at the same time and the previous price changes were also synchronized. This allows the authors to difference between price changes and absorb plant-product-month-age-of-price fixed effects:

$$\Delta_{s_t^{ik}} p_t^{ik} = \theta_{t,s_t^{ik}}^i + \beta_e^k \Delta_{s_t^{ik}} c_t^k + \eta_{t,s_t^{ik}}^{ik}. \quad (4)$$

The fixed effect $\theta_{t,s}^i$ controls for common marginal-cost changes within a plant-product pair. Identification comes from the relative price change between home and UK quote-lines over the same interval.

5.3 Extensive-margin specification

The extensive margin asks whether exchange-rate changes predict asymmetric price changes across markets. Under local-currency invoicing, if firms want to undo the default exchange-rate drift in relative markups, then exchange-rate changes should predict whether UK prices increase or decrease relative to Irish prices.

The paper estimates conditional logit models:

$$\Pr(\text{increase}) = \Lambda \left(\psi_{t,s}^i + (\beta_e^k - 1) \Delta_s e_t^k \right), \quad (5)$$

$$\Pr(\text{decrease}) = \Lambda \left(\phi_{t,s}^i - (\beta_e^k - 1) \Delta_s e_t^k \right). \quad (6)$$

If the coefficients are close to zero, the evidence is consistent with $\beta_e^k = 1$: firms do not systematically undo or amplify the exchange-rate-induced relative markup drift.

6 Main findings

6.1 Finding 1: Prices are sticky, and synchronization is common

Producer prices change infrequently. The full-sample weighted mean frequency of price adjustment is about 0.16 per month, and it is about 0.16 for Sterling-invoiced exports. Price changes across home and UK quote-lines are often synchronized within a plant-product pair, which is essential for the empirical strategy.

6.2 Finding 2: Reset prices show strong pricing-to-market under local-currency invoicing

The intensive-margin estimate is $\widehat{\beta}_e^{UK} = 1.01$, with standard error 0.09. This estimate is statistically different from zero and not statistically different from one. Thus, the relative reset price in the UK moves approximately one-for-one with the exchange rate. Under the assumption of constant relative marginal cost across markets, this means relative markups move one-for-one with exchange rates.

6.3 Finding 3: The result is robust across firms, products, and horizons

The estimate remains close to one across many cuts of the data: plants with different price-adjustment frequencies, different time intervals between price changes, different product types, different plant sizes, different export shares, different UK input shares, different ownership types, and different related-party sales patterns.

6.4 Finding 4: Extensive-margin evidence supports the intensive-margin result

The extensive-margin estimates generally cannot reject $\beta_e^{UK} = 1$. In other words, exchange-rate movements do not systematically predict relative price-change directions in a way that would imply

firms undo the local-currency pricing markup drift.

6.5 Finding 5: Invoice currency matters

For the smaller sample of exports to the UK invoiced in home currency, the evidence is consistent with much weaker pricing-to-market. Under home-currency invoicing, the extensive-margin estimates cannot reject $\beta_e^{UK} = 0$, the opposite of the main local-currency invoicing result.

6.6 Finding 6: The result is not driven simply by failing to control for costs

When the paper estimates an import-pass-through-style regression without plant-product-time fixed effects, the coefficient is about 0.76, similar to findings in Gopinath, Itskhoki, and Rigobon (2010). But the baseline analysis with fixed effects shows that once common marginal-cost movements are controlled for, reset prices imply even stronger pricing-to-market.

7 Interpretation of tables and figures

7.1 Table 1: Summary statistics on plants; Table 2: Weighted mean frequency of price adjustment, invoice currency

Read. Table 1 documents that the PPI sample is more export-oriented and larger than the full CIP plant population. Most exporting plants sell to the UK, and Sterling is the dominant invoice currency for UK sales. Table 2 shows that producer prices are sticky: the weighted mean monthly price-adjustment frequency is 0.16 in the full sample, 0.19 for home-market prices, and 0.16 for Sterling-invoiced exports.

Economic message. The sample is well suited to studying pricing-to-market because UK sales are important, Sterling invoicing is common, and monthly price changes are observed. At the same time, the price stickiness documented in Table 2 makes it crucial to separate mechanical sticky-price effects from reset-price behavior.

TABLE 1
Summary statistics on plants

	1995		2005	
	PPI	CIP	PPI	CIP
% of plants exporting to				
Anywhere	73	53	74	46
UK	67	48	63	39
% of sales exported to				
Anywhere	55	61	75	72
UK	18	16	12	11
% plants invoicing UK sales in				
IEP/EUR only	15	22	24	18
STG only	65	56	44	33
Other only	4	4	2	1
Mix	15	18	30	48
% UK sales invoiced in				
IEP/EUR	11	9	16	20
STG	84	77	70	69
Other	5	13	14	11
% plants importing materials from				
Anywhere	75	58	83	54
UK	69	49	73	48
% of imported intermediates from				
Anywhere	40	48	64	59
UK	16	15	18	16
% plants by ownership				
Foreign	34	16	29	13
UK	7	3	3	2
Employees, distribution across plants				
25th percentile	26	7	22	6
50th percentile	59	15	48	13
75th percentile	129	41	110	36

Notes: PPI refers to the sample of CIP plants that participate in the PPI sample. CIP refers to the full sample of plants. Information on imports is based on the roughly 90% of the population of plants for which comparable information is available for the entire sample period. Information on the export currency is based on the roughly 95% of the population of plants where information is available for the entire sample period.

(a) Table 1: Summary statistics on plants

TABLE 2
Weighted mean frequency of price adjustment, invoice currency

	Unadjusted	Exit adjusted
Full sample	0.16	0.18
By destination market		
Home	0.19	0.20
Export	0.14	0.17
By invoice currency for exports		
IEP/EUR	0.11	0.13
STG	0.16	0.17

Notes: Period covered is January 1995–December 2005. The weighted mean frequency of price adjustment is calculated as $\sum_{i=1}^T \sum_{t=1}^T \omega_i I_t^i / \sum_{i=1}^T \sum_{t=1}^T \omega_i N_t^i$ where I_t^i is an indicator variable equal to 1 if i 's price changed at t , and N_t^i equals 1 if i was present in the sample at t , whether or not its price was changed. ω_i is the plant's sales in the relevant market (home or export) expressed as a share of total within-sample sales in the year corresponding to date t . Exit adjustment treats quote-line exit like a price change, i.e. I_t^i is set equal to 1 if the quote-line is no longer present in the sample at date $t+1$.

(b) Table 2: Weighted mean frequency of price adjustment, invoice currency

7.2 Table 3: Synchronization of price changes within plant-product pairs; Figure 1: IEP/EUR per Pound Sterling over the sample period

Read. Table 3 shows that in the Ireland-UK sample, 20 percent of plant-product-months with more than one quote have at least one price change. Conditional on at least one price change, 28 percent involve only one price changing, 45 percent involve more than one but not all prices changing, and 28 percent involve all prices changing. Figure 1 shows large exchange-rate variation: the Irish pound/euro depreciates substantially against Sterling from the mid-1990s to around 2001, then appreciates after 2002.

Economic message. Table 3 supports the identification strategy because synchronized price changes across markets occur often enough to identify reset-price behavior. Figure 1 supplies the macro variation used to estimate how relative markups respond to exchange-rate movements.

	TABLE 3 Synchronization of price changes within plant-product pairs			
	% of plant-product-months		Of which	
	with > 1 quote and ≥ 1 price change	One price changes	> 1 but < all change	All prices change
Full sample	16	21	28	51
Ireland & UK sample	20	28	45	28

Notes: The first column reports the fraction of plant-product-months with more than one price quote where at least one invoice currency price changes. The second, third, and fourth columns report the fraction of these cases that fall into each of three possible categories. The Irish and UK sample includes plant-product-months with at least one home currency price quote in the home market and at least one Sterling-invoiced price quote in the export market.

(a) Table 3: Synchronization of price changes within plant-product pairs

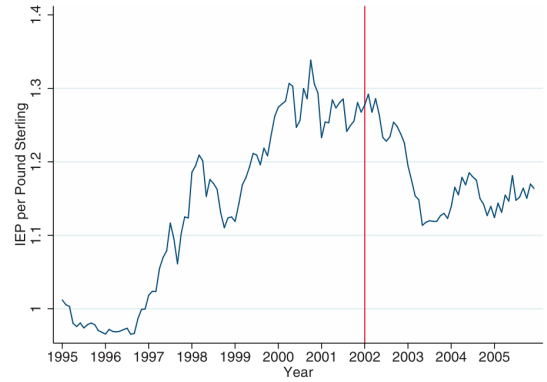


FIGURE 1
IEP/EUR per Pound Sterling over the sample period

(b) Figure 1: IEP/EUR per Pound Sterling over the sample period

7.3 Figure 2: Type of price change episodes used to identify the intensive and extensive margins

Read. Figure 2 illustrates the empirical design. The intensive-margin exercise uses cases where home and UK prices change together at time t , and the previous price changes were also synchronized. The extensive-margin exercise uses cases where previous price changes were synchronized but current price changes are asymmetric across markets or move in different directions.

Economic message. The figure is the cleanest visual explanation of identification. The paper does not use all price changes. It uses carefully selected episodes where common marginal-cost movements can be controlled for by fixed effects.



FIGURE 2
Type of price change episodes used to identify the intensive and extensive margins

Figure 2: Type of price change episodes used to identify the intensive and extensive margins

7.4 Table 4: Intensive margin of price adjustment

Read. Table 4 reports the core OLS intensive-margin estimates. The baseline coefficient on exchange-rate changes is 1.01 with a standard error of 0.09. The coefficient remains close to one across cuts by price-adjustment frequency, interval between synchronized price changes, product type, plant size, UK export share, UK input share, ownership, and related-party sales.

Economic message. This is the central table. It says that, conditional on resetting prices, firms allow the UK-to-Ireland reset-price ratio to move one-for-one with the exchange rate. Under constant relative marginal cost, this is a strong form of pricing-to-market: relative markups move almost exactly with exchange rates.

TABLE 4
Intensive margin of price adjustment

	β_e^{UK}		R^2 -adj.	N	f.e.	clust.
Baseline						
All	1.01	(0.09)**	0.67	4212	1047	86
Median frequency (f) of price adjustment of plant-product pair						
$f < 0.3$	1.07	(0.12)**	0.64	1226	293	68
$0.3 \leq f < 0.5$	0.93	(0.15)**	0.61	1089	313	8
$0.5 \leq f < 0.7$	1.03	(0.33)**	0.67	1269	242	7
$0.7 \geq f$	1.00	(0.39)**	0.69	628	199	5
Time interval between synchronized price changes						
1 mth	0.77	(0.38)**	0.66	748	381	42
2–5 mths	0.93	(0.12)**	0.80	1149	516	46
6–11 mths	0.96	(0.04)**	0.88	1377	578	71
12+ mths	1.03	(0.28)**	0.64	938	441	64
Type of product (Vermeulen <i>et al.</i> (2007))						
Consumer food products	0.91	(0.17)**	0.56	1300	393	20
Cons non-food non-durab	1.86	(0.60)**	0.43	29	13	6
Cons durables	1.06	(0.19)**	0.68	147	28	8
Intermediates	1.11	(0.23)**	0.81	1481	404	31
Capital goods	1.05	(0.10)**	0.60	1251	207	22
Number of employees						
<20	1.25	(0.12)**	0.86	283	98	10
20–29	1.07	(0.23)**	0.67	894	188	24
50–249	1.06	(0.12)**	0.55	2129	477	44
250–499	0.95	(0.17)**	0.61	752	227	7
Quartiles of share of sales exported to the UK						
Q1	0.73	(0.03)**	0.43	68	25	4
Q2	0.94	(0.05)**	0.78	117	25	11
Q3	1.07	(0.22)**	0.58	1865	378	34
Q4	1.01	(0.11)**	0.71	2162	619	57
Quartiles of share of variable cost imported from the UK						
Q1 & Q2	1.03	(0.26)**	0.63	1334	413	32
Q3	1.03	(0.13)**	0.70	617	170	38
Q4	1.00	(0.07)**	0.67	2213	445	40
Ownership						
Domestic	1.04	(0.13)**	0.57	2842	791	64
Foreign	0.96	(0.08)**	0.82	1370	256	24
UK	0.99	(0.01)**	0.96	175	27	8
Share of sales to related parties						
None	1.11	(0.26)**	0.58	1731	466	38
Some	1.11	(0.11)**	0.67	438	111	14

Notes: Estimation method is OLS. Dependent variable is log change in home currency price since last price change. Regressions include a constant and full set of plant-product-month-age-of-price fixed effects. Coefficient on $\Delta_{it} e_t^{UK}$ is reported. Observations are weighted by sales. Standard errors clustered at the plant level in brackets. **signif. diff. from zero at 5% level, *signif diff from zero at 10% level.

Table 4: Intensive margin of price adjustment

7.5 Figure 3: Scatter plot of data used to identify the intensive margin; Figure 4: Size of price and exchange rate changes used to identify the intensive margin

Read. Figure 3 plots the log change in the Sterling UK price against the log change in the euro/Irish-pound home price for synchronized price-change episodes. The observations align around the 45-degree line. Figure 4 compares the distribution of price changes with the distribution of exchange-rate changes used in the intensive-margin estimation.

Economic message. Figure 3 visually supports $\beta_e^{UK} \approx 1$: UK local-currency price changes and Irish home-currency price changes move roughly together when prices reset. Figure 4 shows that price changes are typically larger and more dispersed than exchange-rate changes, so the result is not driven by tiny price adjustments mechanically matching exchange-rate changes.

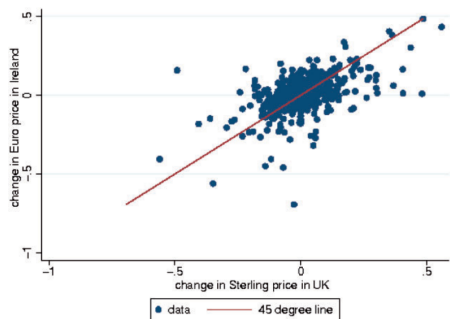


FIGURE 3

Scatter plot of data used to identify the intensive margin

Notes: Figure plots the log change in the Sterling price in the UK against the log change in the home currency price in Ireland. An observation is the change in prices for a plant-product-month where prices change in both markets and where the previous price change was also synchronized across markets. For observations where there is more than one price quote in a market, the mean log change across quotes within that market is plotted.

(a) Figure 3: Scatter plot of data used to identify the intensive margin

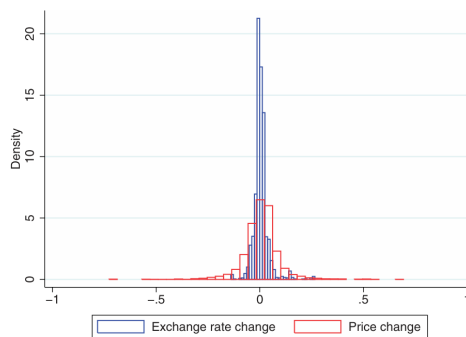


FIGURE 4

Size of price and exchange rate changes used to identify the intensive margin

Notes: Figure plots histograms of the log changes in destination currency prices and the log changes in exchange rates for the sample used in the intensive margin estimation.

(b) Figure 4: Size of price and exchange rate changes used to identify the intensive margin

7.6 Table 5: Extensive margin of price adjustment

Read. Table 5 estimates conditional logit models for the probability of a price increase and the probability of a price decrease. The baseline coefficients are statistically indistinguishable from zero. Across subsamples, estimates are noisy, but there is no systematic evidence that firms use asymmetric price changes to undo the exchange-rate-induced movement in relative markups.

Economic message. The extensive-margin exercise supports the intensive-margin conclusion. The timing and direction of price changes do not suggest that firms systematically offset local-currency-pricing markup drift. This also helps address selection concerns in the intensive-margin estimates.

TABLE 5
Extensive margin of price adjustment

	Probability of a price increase					Probability of a price decrease				
	$\beta_e^{UK}-1$	pseudo- R^2	N	f.e.	clust	$1-\beta_e^{UK}$	pseudo- R^2	N	f.e.	clust
All	0.59 (3.61)	0.00	4873	921	129	1.62 (4.99)	0.00	4564	875	103
Median frequency (f) of price adjustment of plant-product pair										
$f < 0.3$	3.05 (4.79)	0.00	2517	442	111	-5.53 (5.90)	0.00	2187	378	84
$0.3 \leq f < 0.5$	-1.28 (12.05)	0.00	693	172	10	15.73 (7.26)**	0.01	723	187	10
$0.5 \leq f < 0.7$	-0.90 (6.00)	0.00	1160	174	6	5.61 (2.95)	0.00	1129	171	7
$0.7 \geq f$	-13.97 (12.63)	0.00	503	133	4	21.50 (15.00)	0.01	525	139	4
Age of price at second synchronized price change										
1 mth	-4.53 (7.01)	0.00	2537	495	48	8.50 (7.10)	0.00	2424	478	44
2-5 mths	9.47 (8.19)	0.00	1206	232	65	1.70 (8.06)	0.00	1371	258	56
6-11 mths	2.14 (8.26)	0.00	574	98	59	1.67 (9.05)	0.00	450	82	49
12+ mths	0.29 (5.39)	0.00	556	96	62	-4.50 (7.11)	0.01	319	57	40
Type of product (Vermeulen <i>et al.</i> (2007))										
(1)	-1.97 (6.57)	0.00	1408	331	27	-5.35 (6.63)	0.00	1380	321	21
(2)	23.14 (24.79)	0.08	101	26	11	54.75 (47.53)*	0.12	86	16	10
(3)	-1.44 (9.41)	0.00	357	46	14	26.49 (15.11)*	0.08	178	29	10
(4)	1.20 (6.75)	0.00	1904	376	55	3.71 (9.39)	0.00	1840	370	45
(5)	2.20 (6.55)	0.00	1101	141	22	-0.48 (7.99)	0.00	1078	138	19
Plant size										
<20	-12.82 (16.92)	0.01	142	41	13	3.91 (1.89)**	0.00	72	22	7
20-49	4.47 (2.59)*	0.00	1047	164	35	19.96 (10.37)*	0.02	976	155	31
50-249	3.15 (5.07)	0.00	2808	496	68	-2.87 (5.81)	0.00	2586	469	53
250-499	-4.44 (8.03)	0.00	753	187	9	-6.11 (8.15)	0.00	809	196	10
500+	-6.20 (17.20)	0.00	123	33	4	censored to maintain confidentiality				
Quartiles of share of sales exported to the UK										
Q1	-26.65 (18.32)	0.14	108	21	5	-2.98 (9.63)	0.00	95	18	6
Q2	15.03 (27.09)	0.03	117	22	10	-68.26 (17.52)*	0.08	92	17	6
Q3	-2.85 (4.95)	0.00	2054	336	50	4.89 (5.53)	0.00	1866	321	34
Q4	2.91 (5.70)	0.00	2594	542	86	0.04 (7.58)	0.00	2511	519	78
Quartiles of share of variable cost imported from the UK										
Q1&Q2	-5.30 (7.42)	0.00	1354	321	49	9.70 (8.46)	0.01	1465	335	45
Q3	2.18 (6.11)	0.00	907	160	46	-5.07 (6.94)	0.00	769	141	36
Q4	2.46 (5.63)	0.00	2585	429	61	0.20 (7.50)	0.00	2289	382	48
Ownership										
home	0.38 (4.29)	0.00	3356	664	95	5.53 (4.42)	0.00	2999	618	74
foreign	0.89 (6.31)	0.00	1517	257	35	-2.18 (8.79)	0.00	1565	257	31
UK	-7.43 (7.83)	0.02	307	44	8	-14.41 (21.83)	0.01	290	39	8
Share of sales to related parties										
None	-3.98 (3.94)	0.00	1975	392	62	8.44 (3.66)**	0.00	1621	339	45
Some	-5.48 (10.87)	0.00	521	76	19	11.96 (13.42)	0.02	494	75	17

Notes: Estimator is conditional logit, conditioning on plant-product-month-age-of-price fixed effects. Dependent variable is indicator for increase or decrease in invoice currency price. Regressions include a constant. Coefficient on $\Delta_{i,t} \beta_e^{UK}$ is reported. Observations are weighted by sales at the plant-market level. Standard errors clustered at the plant level in brackets. ** signif. diff. from zero at 5% level, * signif diff from zero at 10% level.

Table 5: Extensive margin of price adjustment

7.7 Table 6: Intensive margin by choice of invoice currency; Table 7: Extensive margin by choice of invoice currency

Read. Table 6 compares the intensive margin for local-currency and home-currency invoicing in a smaller sample where UK destination is directly identified. The local-currency coefficient is above one but imprecise; the home-currency coefficient is close to one but also imprecise. Table 7 modifies the extensive-margin exercise by invoice currency. Under local-currency invoicing, the estimates are consistent with $\beta_e^{UK} = 1$; under home-currency invoicing, they are consistent with $\beta_e^{UK} = 0$.

Economic message. Invoice currency is economically meaningful. The main pricing-to-market result is tied to local-currency invoicing. Home-currency invoicing does not produce the same pattern of relative markup movement.

Invoice curr.	β_e^{UK}		R^2 -adj.	N	f.e.	clust
Local	2.52	(1.22)*	0.68	741	235	40
Home	0.97	(0.57)*	0.66	745	236	41

Notes: Sample is restricted to matched pairs of home and foreign sales where destination is identified as the UK in November 2006. Estimation method is OLS. Dependent variable is log change in home currency price since last price change. Regressions include a constant and full set of plant-product-month-age-of-price fixed effects. Coefficient on $\Delta_{p,e}^{UK}$ is reported. Observations are weighted by sales. Standard errors clustered at the plant level in brackets. **signif. diff. from zero at 5% level, *signif diff from zero at 10% level.

(a) Table 6: Intensive margin by choice of invoice currency

	$\Delta_{p,e}^{UK}$		pseudo- R^2	N	f.e.	clust
Local currency invoicing						
Pr(increase)	$\beta_e^{UK} - 1$	10.97	(7.92)	0.00	586	134
Pr(decrease)	$1 - \beta_e^{UK}$	-6.76	(11.57)	0.00	552	131
Home currency invoicing						
Pr(increase)	β_e^{UK}	22.96	(20.17)	0.02	640	151
Pr(decrease)	$-\beta_e^{UK}$	-13.01	(14.57)	0.00	503	129

Notes: Sample is restricted to matched pairs of home and foreign sales where destination is identified as the UK in November 2006. Estimator is conditional logit, conditioning on plant-product-month-age-of-price fixed effects. Dependent variable is indicator for increase or decrease in invoice currency price. Regressions include a constant. Observations are weighted by sales at the plant-market level. Standard errors clustered at the plant level in brackets. **signif. diff. from zero at 5% level, *signif diff from zero at 10% level.

(b) Table 7: Extensive margin by choice of invoice currency

7.8 Table 8: Intensive margin sample robustness; Table 9: Extensive margin sample robustness

Read. Table 8 shows that the intensive-margin result survives several sample restrictions: matching on precise item descriptions, separating cases where delivery charges are included or excluded, expanding to all foreign-currency export sales, and using first-to-last synchronized price changes. Table 9 performs related extensive-margin robustness checks.

Economic message. The results are not an artifact of a loose product match, one narrow market, or delivery-charge contamination. The longer-horizon results are somewhat more suggestive of smaller pricing-to-market beyond two years, but the estimates are imprecise.

	β		R^2 -adj.	N	f.e.	clust
Panel 1						
Matching on precise item description						
All	1.08	(0.16)**	0.80	2169	1054	54
Delivery charges						
Included	1.09	(0.00)**	0.28	58	34	4
Excluded	1.56	(0.62)**	0.42	758	205	30
Panel 2						
All foreign currency export sales						
All	0.99	(0.07)**	0.67	4687	1178	95
Panel 3						
First and last synchronized price changes						
All	0.78	(0.51)	0.39	677	116	96
By interval between first and last synchronized price changes						
<6 months	1.59	(5.03)	0.16	43	23	20
60-11 months	1.38	(0.88)	0.44	65	27	22
120-23 months	1.81	(1.34)	0.47	141	40	35
24+ months	0.74	(0.61)	0.36	428	75	66
By # of intervening price changes						
<2	0.86	(0.46)*	0.33	400	58	49
≥ 2	0.79	(0.66)	0.30	277	58	50

Notes: In Panel 1, results on matching on precise item description are based on defining products by precise item description. In delivery charges regressions, sample includes only quote-lines from 2003 to 2005, present in the November 2006 cross-section, for which the "terms of sale" field for November 2006 reports whether delivery charges are included or excluded from the price. In Panel 2, sample includes all foreign currency export sales. Estimation method is OLS. In Panels 1 and 2, dependent variable is log change in home currency price since last price change. In Panel 3, dependent variable is log change in home currency price between first and last synchronized price change for matched pair of home and UK quote-lines. Regressions include a constant and full set of plant-product-month-difference interval fixed effects. Coefficient on the relevant log exchange rate change is reported. Observations are weighted by sales. Standard errors clustered at the plant level in brackets. **signif. diff. from zero at 5% level, *signif diff from zero at 10% level.

(a) Table 8: Intensive margin sample robustness

Probability of a price increase			Probability of a price decrease						
$\beta - 1$	pseudo- R^2	N	f.e.	clust	$1 - \beta$	pseudo- R^2	N	f.e.	clust
Panel 1									
Matching on precise item description									
All	-2.70	(4.17)	0.00	987	463	70	6.88	(3.30)**	0.01
Delivery charges									
Included	Not enough observations to estimate				Not enough observations to estimate				
Excluded	18.92	(14.86)	0.02	741	36	-8.40	(8.25)	0.00	603
Panel 2									
All foreign currency export sales									
All	2.27	(3.26)	0.00	5458	1052	151	-3.52	(4.16)	0.00

Notes: In Panel 1, results on matching on precise item description are based on defining products by precise item description. In delivery charges regressions, sample includes only quote-lines from 2003 to 2005, present in the November 2006 cross-section, for which the "terms of sale" field for November 2006 reports whether delivery charges are included or excluded in the price. In Panel 2, sample includes all foreign currency export sales. Estimator is conditional logit, conditioning on plant-product-month-age-of-price fixed effects. Dependent variable is indicator for increase or decrease in invoice currency price. Regressions include a constant. Coefficient on the relevant log exchange rate change is reported. Observations are weighted by sales at the plant-market level. Standard errors clustered at the plant level in brackets. **signif. diff. from zero at 5% level, *signif diff from zero at 10% level.

(b) Table 9: Extensive margin sample robustness

7.9 Table 10: Intensive margin, dynamic adjustment; Table 11: Intensive margin, full set of controls

Read. Table 10 adds lagged exchange-rate changes to the intensive-margin regression. The contemporaneous coefficient remains near one, while lagged coefficients are small and insignificant. Table 11 adds destination-market CPI and real-demand controls. The exchange-rate coefficient remains 1.03 with standard error 0.14.

Economic message. The evidence is not driven by delayed pass-through or omitted aggregate demand and price-level variables. The core exchange-rate reset-price relationship survives richer controls.

TABLE 10
Intensive margin: dynamic adjustment

	coeff	s.e.	coeff	s.e.
$\Delta_{it} e_t^{I/K}$	0.93	(0.08)**	0.94	(0.11)**
$\Delta_{it} e_{t-1}^{I/K}$	-0.05	(0.11)	-0.05	(0.12)
$\Delta_{it} e_{t-2}^{I/K}$			-0.17	(0.23)
R^2 -adj.	0.67		0.67	
N	3794		3530	
# f.e.	960		908	
# clusters	67		59	

Notes: Estimation method is OLS. Dependent variable is log change in home currency price since last price change. Regressions include a constant and full set of plant-product-month-age-of-price fixed effects. Observations are weighted by sales. Standard errors clustered at the plant level in brackets. **signif. diff. from zero at 5% level, *signif diff from zero at 10% level.

(a) Table 10: Intensive margin, dynamic adjustment

TABLE 11
Intensive margin: full set of controls

	coeff	s.e.
$\Delta_{it} e_t^{I/K}$	1.03	(0.14)**
$\Delta_{it} p_t^{I/K} X_t^{I/UK}$	-0.45	(1.03)
$\Delta_{it} p_t^{I/K} X_t^{I/UK}$	0.73	(0.90)
$\Delta_{it} q_t^{I/K} X_t^{I/UK}$	0.82	(0.76)
$\Delta_{it} p_t^{I/K} X_t^{I/UK}$	-1.36	(1.19)
R^2 -adj.	0.66	
N	3325	
# f.e.	802	
# clusters	75	

Notes: Sample covers 1997–2005. Estimation method is OLS. Dependent variable is log change in home currency price since last price change. Regression includes a constant, $X_t^{I/UK}$, an indicator variable for the Irish market and full set of plant-product-month-age-of-price fixed effects. Observations are weighted by sales. Standard errors clustered at the plant level in brackets. **signif. diff. from zero at 5% level, *signif diff from zero at 10% level.

(b) Table 11: Intensive margin, full set of controls

7.10 Table 12: Extensive margin, full set of controls; Table 13: Intensive margin, not controlling for costs

Read. Table 12 repeats the extensive-margin exercise with the full set of controls. The exchange-rate coefficient remains statistically insignificant in both the price-increase and price-decrease equations. Table 13 estimates an intensive-margin regression without controlling for plant-product-time costs, using quote-line fixed effects instead. The coefficient is 0.76 with standard error 0.14.

Economic message. Table 12 reinforces the extensive-margin result. Table 13 shows why the paper’s cost-control strategy matters: without controlling for common marginal cost, the data produce a pass-through-style coefficient similar to the import-price literature; with cost controls, the evidence points to a stronger markup-adjustment interpretation.

TABLE 12
Extensive margin: full set of controls

	Increases		Decreases	
	coeff	s.e.	coeff	s.e.
$\Delta_{it} e_t^{I/K}$	5.78	(4.46)	0.10	(5.39)
$\Delta_{it} p_t^{I/K} X_t^{I/UK}$	26.01	(21.02)	-13.48	(24.83)
$\Delta_{it} p_t^{I/K} X_t^{I/UK}$	19.47	(23.82)	-6.07	(21.85)
$\Delta_{it} q_t^{I/K} X_t^{I/UK}$	-18.13	(16.46)	38.80	(18.66)**
$\Delta_{it} p_t^{I/K} X_t^{I/UK}$	30.12	(22.61)	-57.79	(26.24)**
Pseudo- R^2	0.01		0.01	
N	3899		3711	
# f.e.	719		688	
# clusters	112		94	

Notes: Sample covers 1997–2005. Estimator is conditional logit, conditioning on plant-product-month-age-of-price fixed effects. Dependent variable is indicator for increase or decrease in invoice currency price. Regressions include a constant and $X_t^{I/UK}$, an indicator for the Irish market. Observations are weighted by sales. Standard errors clustered at the plant level in brackets. **signif. diff. from zero at 5% level, *signif diff from zero at 10% level.

(a) Table 12: Extensive margin, full set of controls

TABLE 13					
<i>Intensive margin, not controlling for costs</i>					
$\Delta_{it} e_t^{I/K}$		R^2 -adj.	N	f.e.	clust
0.76	(0.14)**	0.15	5190	600	162

Notes: Sample includes all Sterling-invoiced quote-lines. Estimation method is OLS. Dependent variable is log change in home currency price since last price change. Regression includes a constant and full set of quote-line fixed effects. Observations are weighted by sales at the level of the plant-market. Standard errors clustered at the plant level in brackets. **signif. diff. from zero at 5% level, *signif diff from zero at 10% level.

(b) Table 13: Intensive margin, not controlling for costs

8 Identification assumptions and limitations

Constant relative marginal cost. The key identifying assumption is that relative marginal cost across the Irish and UK markets is constant within a plant-product pair. This lets changes in relative reset prices reveal changes in relative markups. The factory-gate nature of the data and same-plant same-product comparisons make this assumption more credible than in aggregate export data.

Product matching. Product categories may not always perfectly identify identical products. The paper addresses this by using stricter item-description matches in robustness checks. Results are similar.

Delivery charges. If delivery charges are included differently across home and export prices, relative marginal cost could move with exchange rates. The paper checks subsamples where delivery charges are reported as included or excluded, and the results do not change systematically.

Selection into price changes. The intensive-margin estimation observes reset prices only when prices change, and it further requires synchronized price changes. This could create selection bias. The extensive-margin exercise is less exposed to this issue and yields similar conclusions.

Destination-market identification. For most of the sample, export destination is not always directly observed. The main analysis therefore focuses on Sterling-invoiced exports, which are very likely UK sales. This is a strength for identifying the UK market but limits the baseline sample.

9 How to connect this paper to the literature

The paper contributes to three literatures.

Pricing-to-market and exchange-rate pass-through. It builds on Knetter-style fixed-effect methods but uses much cleaner microdata: same plant, same product category, home and export prices, factory-gate observations, invoice currency, and timing of price changes.

Sticky prices and reset prices. The paper relates to the reset-price literature by showing that reset prices themselves respond to exchange rates in a way that sustains real exchange-rate movements.

International macro and real exchange-rate puzzles. Standard sticky-price models have difficulty generating enough real exchange-rate persistence. The paper suggests that if reset prices move with exchange rates, not merely stale prices, the real exchange rate can remain volatile and persistent even when price changes occur.

10 Short summary

Fitzgerald and Haller study Irish manufacturing plants that sell the same product domestically and to the UK. The data allow them to observe monthly factory-gate prices, invoice currency, and price-change timing. Their main empirical strategy compares synchronized reset-price changes across Irish and UK markets, using plant-product-month-age-of-price fixed effects to absorb common marginal-cost movements. They find that for Sterling-invoiced UK export prices, the relative reset price in the UK moves almost one-for-one with the nominal exchange rate. Under constant relative marginal costs, this means relative markups move one-for-one with exchange rates. The result is robust across products, plants, and specifications, and it is supported by extensive-margin evidence on asymmetric price changes.

11 Conclusion and takeaway

The paper's main takeaway is that pricing-to-market is not only a mechanical implication of sticky local-currency prices. Even when firms reset prices, they choose prices that allow foreign-market markups to move strongly with exchange rates. This is a powerful empirical fact for international macro and trade pricing models. It says that models should not treat reset prices as automatically realigning relative prices across markets. Instead, realistic reset-price behavior may itself be an important source of real exchange-rate persistence and volatility.